



In addition to the part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course –

Course No. : ME G552

Course Title : Industrial Robotics and Controls

Instructor-in-Charge : Dr. Yogesh Singh (yogesh.singh@pilani.bits-pilani.ac.in)

1. Course Description:

This course provides a comprehensive foundation in industrial robotic systems and robot control. Topics include principles of robot kinematics, statics, dynamics, and control; rigid body motion and coordinate transformations; rotation matrices, Denavit-Hartenberg parameters forward and inverse kinematics; Jacobian analysis, workspace and singularity analysis; trajectory planning and motion optimization; robot statics and dynamics using Euler–Lagrange and Newton–Euler formulations; control system for robot, independent joint control, computed torque control, and force control; end-effector and gripper design for robotic manipulation and precision assembly; sensor-integrated robotic systems; collaborative robotics; robotic work-cell implementation; and case studies and applications of industrial robots in intelligent manufacturing environments.

2. Scope and Objective of the Course:

This course provides fundamental, theoretical, and applied knowledge of industrial robotics and robot control with a strong emphasis on hands-on learning and industry-oriented applications. The course develops a rigorous foundation in robot kinematics, dynamics, motion planning, and control required for the modelling, analysis, and operation of industrial robotic manipulators.

Students will gain practical exposure to industrial robotic systems used in modern manufacturing environments for robotic manipulation, precision assembly, collaborative robotics, and intelligent production systems. Laboratory sessions and project-based learning will focus on robotic manipulator programming, trajectory generation, end-effector design, sensor-integrated robotic systems, and robotic work-cell implementation using industrial-grade platforms.

Course Outcome:

Upon successful completion of this course, students will be able to:

CO1	Apply principles of robot kinematics, dynamics, and control to model and analyze industrial robotic manipulators.
CO2	Design and implement motion planning and control strategies for industrial robotic tasks such as pick-and-place, assembly, and material handling.
CO3	Design and deploy sensor-integrated robotic work cells for real-world manufacturing applications with closed-loop control.
CO4	Demonstrate hands-on proficiency and industry readiness by solving practical robotics problems through labs, projects, and industry-led sessions.





3. Textbook(s):

(T1) Introduction to Robotics: Mechanics and Control — John J. Craig
For: Fundamental concepts of robot kinematics, dynamics, and control.

(T2) Robotics: Modelling, Planning and Control — Bruno Siciliano et al.
For: Advanced mathematical treatment of robotics modeling and planning.

4. Reference Book(s):

(R1) Lung-Wen Tsai, “Robot Analysis – The mechanics of serial and parallel Manipulators”, John Wiley & Sons, Inc

(R2) Mittal R. K. & Nagrath, I. J., “Robotics and Control”, TMH, 2003 (Reprint 2010 or later)

(R3) Mark W. Spong and M. Vidyasagar, “Robot, Dynamics, and Control”, John Wiley & Sons, Inc.

5. Lecture Plan:

Module #	Lecture Session	Reference	Learning Outcome
M1	Robot Terminology (M1) Lectures 1 to 8 – Introduction to Industrial Robotics, Mechanisms and Machines, Types of Joints and Degree of Freedoms, Pose of a Rigid Body, Rotation Matrices, Euler Angles, Quaternions, Homogenous Transformations, Denavit Hartenberg Convention and Parameters.	T1	Analyze and apply concepts of rigid body motion, coordinate transformations, and Denavit-Hartenberg representation to systematically model industrial robotic manipulators
M2	Robot Kinematics and Trajectory Planning (M2) Lectures 9 to 20 – Forward and Inverse Kinematics, Jacobian, Workspace Analysis, Singularity in a Robot, Trajectory Planning, Manipulability, Velocity Ellipses.	T1, R1	Analyze and solve forward and inverse kinematics problems, evaluate workspace and singularities, and design trajectory planning strategies for efficient robotic motion.
M3	Robot Statics and Dynamics (M3) Lectures 21 to 28 – Statics and Force/Torque Ellipses, Dynamics using Euler-Lagrange equations and Newton-Euler Formulation.	T1, T2, R1	Formulate, analyze, and evaluate static and dynamic models of robotic systems using Newton-Euler and Lagrangian methods to determine force and torque requirements.
M4	Robot Control (M4) Lectures 29 – 36 – Open and Closed Loop System, Independent Joint Control, Computed Joint Torque/Multivariate Control, Force Control.	T1, R2, R3	Design and evaluate advanced robot control strategies, including independent joint control, computed torque control, and force control, for stable and precise manipulation.
M5	Robot End-Effector Design (M5) Lectures 36 – 42 – Robot Pick and Place operations with end-effector design using the concepts learnt.	Lecture Notes	Design and evaluate end-effectors and tooling systems for industrial applications by synthesizing task requirements, constraints, and operational





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ACADEMIC GRADUATE STUDIES AND RESEARCH DIVISION

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6. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Lab Practicals/Hands-On Experiments	30	TBA	Post each module	- Perform in a team of maximum 3 students. - Best 4 of 5 - Take home
Case Studies	15	TBA	Towards end of semester	Perform in a same team as above
Mid Semester Exam	20	90 min(s)	As announced in the timetable	Open Book
Comprehensive Exam	35	180 min(s)	As announced in the timetable	Open Book and/or partly Software Based (MATLAB, Python, Robot Software, etc.)

Note - Minimum Passing Marks Criteria:

The passing marks for this course will be determined based on one of the criteria - 30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as a passing mark in this course. Failure to meet this criterion will result in an NC grade.

7. Office Consultation Hour:

Announced or by prior appointment via email. Reach out to the IC at yogesh.singh@pilani.bits-pilani.ac.in
Office Location: NAB 6020-D.

8. Make-up Policy

Make-up will be granted only in genuine cases with prior permission as per the institute's rules.

9. Attendance Policy

Attendance is not mandatory. However, students are strongly encouraged to attend lectures, actively engage in class discussions, and benefit from the learning experience.

Instructor-In-Charge
ME G552: Industrial Robotics and Control



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